

Sundew Dx

Objective

Enable **early, low-cost, non-invasive, and accurate detection of lung cancer** by analyzing volatile organic compounds in exhaled breath condensate **using RNA aptasensing technology** to increase rates of early lung cancer detection.

Target Market

- TAM: **\$15.1 Billion (2023)** to a projected **\$34.8 Billion (2034)**
- SAM: **US - \$4.4 Billion (2023)** to a projected **\$7.6 Billion (2030)**
- SOM: Non-invasive early detection in the US - **\$115 Million**

Value Proposition

This tool has potential to be deployed in **point-of-care settings to enable earlier diagnosis**, improve survival rates, reduce healthcare costs, and **increase accessibility**, especially in underserved populations. There is prospective to adapt the core technology for other disease biomarkers/cases.

Business Model (SPACED)

Relies on **device sales plus recurring revenue from disposable sensor cartridges**, supported by AI-driven analytics for accuracy. Customers would include individuals, hospitals, screening programs, insurers, and eventually pharmacies or employers for broader access. Success hinges on **clinical validation, regulatory approval, and strategic partnerships** with healthcare providers, insurers, and manufacturers.

Financial Projections and Near-Term Cash Needs

Immediate rounds of capital will be applied to R&D of aptamer selection and biosensing fine-tuning:

- Aptamer selection: **\$100k** with a minimum of 50 rounds of enrichment
- Target completion: **1-2 years from date of investment (DOI)**

Funds will then be placed on multiple rounds of semiconductor identification and optimization for this use case:

- Semiconductor identification: **\$375k** with multiple testing rounds
- Target completion: 1-3 years from DOI

Investment Goal: \$2.25 million over 3 years

- **Year 1 (\$500,000):** Aptamer development, semiconductor identification, lab-on-a-chip prototyping, initial talent acquisition
- **Year 2 (\$1,000,000):** Semiconductor optimization, breath capture integration, usability testing, synthetic breath model validation
- **Year 3 (\$750,000):** scalable manufacturing, market preparation, clinical

Investor Value Beyond Capital

- **Strategic input:** Help shape commercialization roadmap and scaling strategy
- **Regulatory expertise:** Support FDA navigation and de-risk the submission process, if necessary
- **Exit planning:** Guide toward acquisition or IPO via C-Corp framework
- **Network access:** Open doors to talent, clinical data partners, manufacturing resources, and distribution channels